

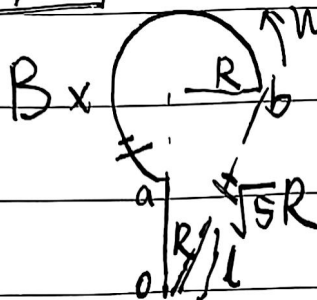
14.3 $\Phi = \vec{B} \cdot \vec{S}$

动力生: $\vec{S}$ 变化导致 Φ 变化

感生: $\vec{B}$ 变化导致 Φ 变化

14.3

连接 aob , $\Sigma_{aob} = \Sigma_{ob}$

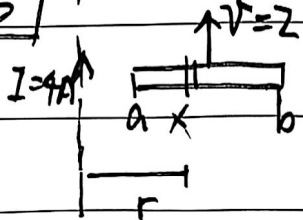


$$d\Sigma_{ob} = w l B dl$$

$$\Sigma_{ob} = \int_0^{5R} w B l dl = \frac{5wB}{2} R^2$$

$$\varphi_a > \varphi_b$$

14.5



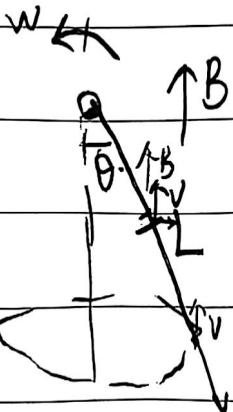
$$B(r) = \frac{\mu_0 I}{2\pi r}$$

$$d\Sigma = B(r) v dr$$

$$\Sigma = \int_{a_1}^{a_2} \frac{\mu_0 I}{2\pi} v \frac{dr}{r} = \frac{\mu_0 I}{2\pi} v \ln 10 = 3.7 \times 10^{-6} v$$

a 端高

14.6



$$d\Sigma = (\vec{v} \times \vec{B}) dL = v B \sin\theta dL = B w \sin^2\theta L dL$$

$$\Sigma = \int_0^L d\Sigma = \frac{B w \sin^2\theta}{2} L^2$$

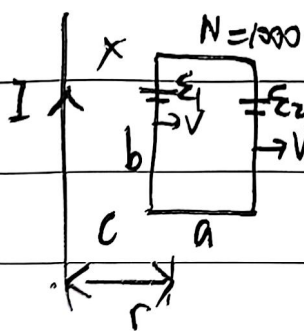
方向如图

14.7

$$B(r) = \frac{\mu_0 I}{2\pi r}$$

$$\frac{1}{c} - \frac{1}{a+c}$$

$$\frac{a}{c(a+c)}$$



$$\Sigma_1 = v B(a) b = N \frac{v \mu_0 I}{2\pi} \frac{b}{c}$$

$$\Sigma_2 = v B(a+c) b = N \frac{v \mu_0 I}{2\pi} \frac{b}{a+c}$$

$$\Sigma = \Sigma_1 - \Sigma_2 = v B a b = \frac{N v \mu_0 I b}{2\pi} \left(\frac{a}{c(a+c)} \right)$$

$$= 6.9 \times 10^{-6} v$$

14.9) $B = 4t^2 y$ $l = 0.2m$

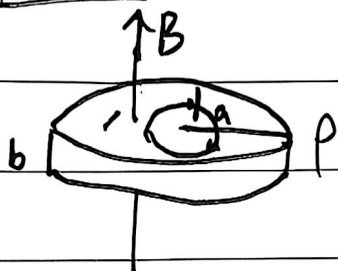
$$|\mathcal{E}| = \frac{d\Phi}{dt}$$

$$\Phi = \int B \cdot dS = \int_0^l B \cdot l \cdot dy = \int_0^l 4t^2 l y dy = 2l^3 t^2$$

$$\therefore |\mathcal{E}| = 4l^3 t \Big|_{t=0.25} = 0.008 V \quad \text{|| 顺时针 ||}$$

14.11

设 $\vec{B}$ 竖直向上



$$\Sigma = -\frac{d\Phi}{dt} = -\frac{dB}{dt} \cdot \pi a^2 = -k\pi a^2 \cdot r^2$$

$$\frac{d\vec{i}}{dr} = \frac{\Sigma}{\rho \frac{2\pi r}{b} dr} = -\frac{k\pi a^2 b dr}{2\rho \cdot r}$$

$$\therefore \vec{i} = \int_0^a d\vec{i} = -\frac{ka^2 b}{2\rho}$$

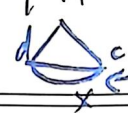
理解本质: $\Sigma = -\frac{d\Phi}{dt}$ 是对“线”讲的, 不是“面”

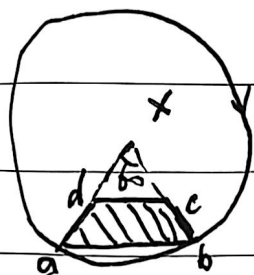
$$\Sigma(r) = -k\pi r^2$$

$$d\vec{i} = \frac{\Sigma}{\rho \frac{2\pi r}{b} dr} = -\frac{kb}{2\rho} r dr$$

$$\vec{i} = \int_0^a -\frac{kb}{2\rho} r dr = -\frac{kb}{4\rho} a^2$$

14.15

(1) 这是  将cd等效为Σa是错误的



$$\Sigma da = \Sigma bc = 0$$

$$\Sigma = -\frac{d\Phi}{dt} \dots$$

对cd: $E_1 \cdot 2\pi \cdot \frac{R}{2} = -\frac{dB}{dt} \cdot \pi \left(\frac{R}{2}\right)^2 \Rightarrow E_1 = -\frac{R}{4} \frac{dB}{dt}$

$$\Sigma_{cd} = E_1 \cdot \frac{\pi R}{6} = -\frac{\pi R^2}{24} \frac{dB}{dt}$$

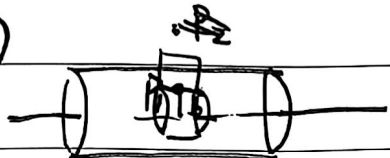
对ab: $E_2 \cdot 2\pi R = -\frac{dB}{dt} \cdot \pi R^2 \Rightarrow E_2 = -\frac{R}{2} \frac{dB}{dt}$

$$\Sigma_{ab} = -E_2 \cdot \frac{\pi R}{3} = \frac{\pi R^2}{6} \frac{dB}{dt}$$

(2) $\Sigma_{\text{总}} = \Sigma_{ab} + \Sigma_{bc} + \Sigma_{cd} + \Sigma_{da} = \frac{\pi R^2}{8} \frac{dB}{dt}$ 逆时针

14.17. λ

①



~~$$E \cdot 2\pi b \cdot dl = \frac{1}{\epsilon_0} \cdot 0 \Rightarrow E_{P1} = 0$$~~

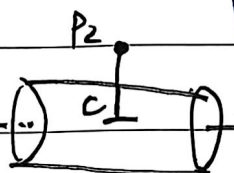
$$E \cdot 2\pi b = -\frac{d\Phi}{dt}$$

② $B \cdot dl = \mu_0 \cdot I$, $I = \lambda dl \frac{v}{2\pi} \Rightarrow B = \frac{\mu_0 \lambda v}{2\pi}$

~~$$\Phi = B \cdot \pi b^2$$~~
$$\frac{d\Phi}{dt} = \pi b^2 \frac{\mu_0 \lambda}{2\pi} \beta = \frac{\mu_0 \lambda b^2}{2} \beta$$

$$\therefore E = -\frac{\mu_0 \lambda b \beta}{4\pi}$$

②



$$E \cdot 2\pi c = -\frac{d\Phi}{dt}$$

$$\Phi = B \cdot \pi a^2 \quad \frac{d\Phi}{dt} = \frac{\mu_0 \lambda a^2}{2} \beta$$

$$\therefore E = -\frac{\mu_0 \lambda a^2 \beta}{4\pi c}$$

$$\therefore |\Sigma_{mv}| = |E \cdot 2\pi \arctan \frac{1}{2c} \cdot \sqrt{c^2 + \frac{1}{4}}| = \frac{\mu_0 \lambda a^2 \beta}{2\pi c} \left(\arctan \frac{1}{2c} \right) \sqrt{c^2 + \frac{1}{4}}$$

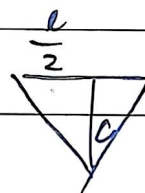
~~$$\Sigma = -\frac{d\Phi}{dt}$$~~

$$\Sigma = -\frac{d\Phi}{dt}$$

$$\Phi = \frac{\mu_0 \lambda a^2 v}{2} \cdot \frac{\theta}{2\pi}$$

$$\theta = 2\pi \arctan \frac{1}{c}$$

$$\Sigma = -\frac{\mu_0 \lambda a^2 \beta}{2\pi} \arctan \frac{1}{2c}$$



同样的错误